MILITARY SPECIFICATION

RADIO SET, AN/PRC-125
(NAVY SWIMMER RADIO)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. This specification establishes the performance, design, manufacture, and testing requirements of the Radio Set, AN/PRC-125 (Navy Swimmer Radio). This radio set is designed for use by rescue swimmer operating from Navy shore installations and aircraft. The radio set shall be of modular design and construction as specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warefare Center, Aircraft Division Lakehurst, Systems Requirement Department Code SR3 Lakehurst NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-5821

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2.0 APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified in the solicitation, the following specifications and standards form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-F-325	Filler, Engraving, Stamped Marking

Military

MIL-B-18/267	Battery, Dry BA-1568/U
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-P-19834	Plates, Identification, Metal Foil Adhesive Backed and Associated Equipment
MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic
MIL-C-22442	Cable Assemblied Radio Frequency (AIRCRAFT AUDIO) General Specification For
MIL-M-38510	Microcircuits General Specification For

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Electronic Equipment, General Requirements for

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MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for Control of Electromagnetic Interference
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-794	Preservation, Packaging, Packing and Marking
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	Systems Safety Program for Systems and Associated Subsystems and Equipment Requirements for
MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2074	Failure Classification for Reliability Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5049.

2.1.2 Other Government documents, drawings and publications. The following other Government documents , drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DL1880AS100 Radio AN/PRC 125

(Copies of DL1880AS100 may be obtained from the Naval Air Warfare Center Aircraft Division Indianapolis, 6000 East 21st Street, Indianapolis IN 46219-2189.)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1141-52 Ocean Water Substitution, Specification for

Application for copies of D1141-52 should be addressed to ASTM, 1916 Race Street, Philadelphia, PA, 19103.

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3.0 REQUIREMENTS

3.1 First Article. First Article tests shall be as specified by the contract for First Article inspection. The samples that shall be subjected to First Article inspection shall be the tests specified in paragraph 4.6 herein.

3.2 Parts, Materials and Processes

3.2.1 Selection of Parts and Materials. Selection of parts and materials shall be in accordance with MIL-E-5400 class 1B.

3.2.2 Microelectronic Devices. Microelectronic devices shall be used to the maximum extent possible in the radio set. Selection of microelectronic devices shall be in accordance with Requirement 64 of MIL-STD-454. All hybrid circuits shall conform to the requirements of MIL-M-38510.

3.3 Design and Construction. The radio sets shall conform with all applicable requirements of MIL-E-5400, class 1B for design, construction, workmanship and safety. The radio sets shall meet all the requirements of this document and the Naval Air Systems Command techinical data package 1880AS100. (See Figure 1)

3.3.1 Operation. The radio set shall remain in the receive mode until the push-to-talk (PTT) switch activates the transmit mode. The radio set shall contain operator selected channels of 243.0 MHz and 282.8 MHz. In addition, an operator selected beacon mode will be provided on 243.0 MHZ.

3.3.2 Type of Transmission. The transmitter shall provide Amplitude Modulated (AM) emission (A3).

3.3.3 Reliability. The reliability of the radio set shall be determined in accordance with the requirements of MIL-STD-781.

3.3.3.1 Mean Time Between Failures (MTBF). The Mean Time Between Failures of the radio set shall have a MTBF (01) of 750 hours for the lower test and a MTBF (01) of 1500 hours for the upper test in accordance with MIL-STD-781.

3.3.3.2 Reliability Testing. The AN/PRC-125 Radio Set (not including battery) shall meet the performance requirements of paragraph 3.5 after being subjected to reliability testing specified in 4.9.28.

3.3.2 Interchangeability. Physical and functional interchangeability shall exist for all radio sets and replaceable assemblies, subassemblies, and parts of the radio sets, in accordance with MIL-E-5400 class 1B. The radio sets shall be constructed to preclude the requirement for any adjustment when replacing a unit.

3.3.3 Reversibility. The equipment shall incorporate features such that it is mechanically or electrically impossible to incorrectly install any printed wiring cards or subassemblies. Radio Frequency (RF) connectors shall be legibly marked with reference designators in accordance with MIL-E-5400 class 1b.

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3.3.4 Supportability.

3.3.4.1 Service and Access. The radio set is of modular design, consisting of the R/T unit and control unit/antenna, with attached umbilical. Each module is a sealed unit. Organizational level maintenance, is the only maintenance required.

3.3.4.2 Nonrepairable Assemblies. The AN/PRC-125 Radio Set is designed as nonrepairable as defined in paragraph 3.2 of MIL-E-5400 class 1B.

3.3.5 Dimensions. Excluding switches, connectors, and antenna, the radio set dimensions shall not exceed the following:

3.3.5.1 Control Unit.

a.	Depth	1.4 inches
b.	Width	3.0 inches
c.	Height	3.4 inches

3.3.5.2 Receiver/Transmitter Unit.

a.	Depth	1.4 inches
b.	Width	3.1 inches
c.	Height	6.0 inches

3.3.6 Weight. The weight of the radio set, including the control unit and antenna but excluding battery, shall not be greater than 2.25 pounds.

3.3.7 Power Source. The radio set shall be powered by a battery designated as Dry BA-1568/U, in accordance with MIL-B-18/267, The battery compartment shall be separately sealed from the remainder of the R/T to provide watertight integrity. The battery shall be replaceable with out the use of tools.

3.3.7.1 Battery Reversal. The battery compartment shall be designed so that it is physically impossible to insert the battery incorrectly.

3.3.7.2 Battery Operating Voltage and Current. The radio set shall provide specified performance over the limits of direct current (DC) supply power specified below:

a. Transmit and receive shall be 10.0 to 13.0 volts DC.

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b. The DC input current shall not be greater than 100 mA at 10.0 volts DC and 20.0 dBm average Root Mean Square (RMS) RF power output, unmodulated in the transmit mode. (dBm referenced to 1.0 milliwatt, 20 dBm is equivalent to 100 milliwatt) Input current shall not be greater than 80 mA in the receive mode with a 10 hard microvolt unmodulated signal. (One "hard" microvolt is defined as the RF signal at the input of a receiver when the receiver is connected through a 50 ohm, 6 dB attenuator, to a signal generator, with its output set at 10.0 microvolts.)

c. It shall be acceptable for the transmit mode on either channel to have a higher RF power output; however, the maximum input current in the transmit mode, unmodulated, shall be 105 mA at 10.0 volts DC.

3.3.8 Manual operating Controls. The manually operated controls shall be as specified in paragraphs 3.4.9, 3.4.10, and 3.4.11 herein.

3.3.8.1 Frequency Selector. The frequency selector shall be a four position switch allowing operator selection of the mode. The lowest position shall be marked 282.8 MHz. The next upward position shall be marked OFF; in this mode all power to the transceiver shall be removed. The next upward position shall be marked 243.0 MHz. The most upward position shall be marked BCN (Beacon). It shall not be possible to select either switch position without first depressing the frequency selector control.

3.3.8.2 Volume Control. The volume control shall operate in the receive mode. The receiver volume shall be adjusted by means of a single control. Frictional drag shall be provided to help preclude inadvertent adjustment of the control.

3.3.8.3 Push to Talk Switch. The Push to Talk Switch (PTT) shall be integrated into the control unit. Depressing the PTT switch shall de-energize the receiver and activate the transmit mode, giving transmitter operation on the channel selected by the user. The transmit mode shall activate before the PTT switch becomes flush with the control unit case. It shall not be necessary to press the PTT switch to activate the BCN.

3.3.9 Control Unit. The control unit shall consist of an umbilical cable with strain relief, pressure relief valve, speaker/microphone, earphone connector, PTT switch, volume control, and any impedance matching or control circuitry necessary to perform all required functions. A single transducer may be used to perform both speaker and microphone functions provided all requirements are met. The control unit is used to mount the antenna see 3.3.3.13. The entire assembly shall be suitably waterproofed to meet the immersion requirements specified herein.

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3.3.9.1 Control Unit Functions. The speaker shall be connected to the receiver output at all times until the PTT switch activates the transmit mode. In the transmit mode the speaker shall be disengaged and the microphone shall be connected to the transmitter input. The control unit shall contain circuity to provide the appropriate impedance matches between the speaker/microphone and the R/T input and output, and between the R/T output and the external earphone when the earphone is connected see 3.3.3.11. The control unit shall contain a means of controlling the speaker volume.

3.3.10 Umbilical Cable. An umbilical cable, sealed to withstand the specified immersion requirements, shall connect the control unit to the R/T housing. The RF cable shall be part of the umbilical cable. The umbilical cable shall be strain relieved and removeable at the R/T end. The cable assembly shall be in accordance with the following requirements and tests of MIL-C-22442.

a. Dielectric Withstanding Voltage

b. Gauging

c. Pull Test

d. Bend Relief Flexibility

e. Low Temperature Flexibility

f. Durability

g. Ozone Resistance

h. Jacket Elongation

3.3.11 Earphone Characteristics. The radio set shall be designed to operate with an earphone furnished in accordance with Specification Control Drawing 12-520447. A connection shall be provided on the control unit for use with the earphone. Damage to the internal circuits shall not occur when the earphone connection is terminated in any load from short to open circuit.

3.3.12 Attachment Features. The control unit shall be equipped for attachment, at the back of the control unit, to the Model LPU-28/P life vest. The method of attachment shall be hook and pile material in accordance with MIL-F-21840 Type II, Class 1 Black 3230.

3.3.13 Antenna. The antenna shall be mounted on top of the control unit and supplied as part of the radio set as a separate, replaceable unit or assembly.

3.3.13.1 Voltage Standing Wave Ratio (VSWR). The VSWR of the antenna and control unit combination shall not be greater than 2.0:1 at 243.0 MHz and 282.8 MHz.

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3.3.13.2 Radiation Pattern. The pattern produced by the antenna shall be horizontal, and omidirectional. The antenna shall be base fed and vertically polarized.

3.3.13.3 Mechanical Features. The antenna design shall not cause injury to the swimmer during operation (including water entry). The antenna shall be suitable to provide underwater survivability to a depth of 30 feet.

3.3.14 Receiver sensitivity shall comply with paragraph 3.6.3.2.3., when the SDU5/E strobe light is operated within 18 inches of the operating radio set.

3.3.15 Casing Finish. The outside cases of the R/T unit and control unit shall be finished with a black anodize per MIL-A-8625 Type II, Class 2, or olive drab enamel in accordance with FED-STD-595, Color 34087.

3.3.16 Case Lettering. Case lettering shall be recessed and filled with Filler Crayon Type 1, per TT-F-325, color White per FED-STD-595, color No. 37778.

3.3.17 Nameplates and Identification Marking. Serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Identification marking shall be in accordance with MIL-STD-130.

3.3.17.1 Nameplate. A metal foil nameplate, in accordance with MIL-P-19834, Type I, Style III, shall be permanently affixed to the radio set. If necessary to prevent corrosion, the nameplate shall be sealed.

3.3.17.2 Serial Numbers. Serial numbers and prefixes shall be assigned by the procuring activity, for the radio set under contract or purchase order. The contractor shall serialize sequentially all radio sets which are delivered under contract.

3.3.18 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-STD-882.

3.4 Performance.

3.4.1 Stress Screening (Burn-in). Each radio set shall be subject to stress screening (burn in) prior commencement of any tests or submission of the unit for acceptance. The burn-in shall be performed on all radio sets and the radio sets shall be operational during the burn-in period. A minimum of two temperature cycles will be run on each radio set with the last four hours failure free.

3.4.2 Sidetone. A transmitter sidetone shall be provided to the earphone terminals, when the earphone is connected, and the radio set is in the transmit mode. The sidetone shall be active only when the earphone is connected. The level of the sidetone shall not be affected by the position of the volume control and shall be of sufficient level to be audible to the user when the ambient noise level is 102 dB SPL.

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3.4.3 Warm-up Time. The warm-up time required prior to operation shall not be greater than 1.0 second.

3.4.4 Transmitter Characteristics

3.4.4.1 Frequency Range. The transmitter shall be capable of providing voice communications on either of two preset frequencies, one of which shall be 243.0 MHz, the other shall be 282.8 MHz. The BCN mode shall operate on 243.0 MHz only.

3.4.4.2 Frequency Stability and Accuracy. The frequency of the transmitter shall not deviate by more than +/- 0.005 percent from the selected frequency.

3.4.4.3 Microphone Sensitivity. Sensitivity of the microphone shall be sufficient to provide the required audio modulation of the R/T in the transmit mode as specified in 3.4.4.1, with a sound source of 102 dB SPL located 1.0 inch from the face of the microphone.

3.4.4.4 Modulation. The modulation capability shall be 80 percent +/- 10%. The modulation source shall be pink noise, equivalent in amplitude to that produced by the microphone, with a sound source of 102 dB SPL, located 1.0 inch from the face of the microphone. Operating conditions shall be standard, with an input voltage of 10.0 volt DC. Under no circumstance shall the modulation be greater than 100 percent.

3.4.4.5 Distortion. With 80 percent sine wave modulation, in the range between 500 and 3000 Hz, the Total Harmonic Distortion (THD) of the audio envelope shall not be greater than 15 percent.

3.4.4.6 Fidelity. Overall transmitter fidelity of the amplifier and modulation circuits, excluding the microphone response, for sine wave inputs in the range between 500 and 3000 Hz shall be within +/- 3.0 dB of the response at 1000 Hz.

3.4.4.7 RF Power Output. The average unmodulated RMS continuous wave RF power output of the transmitter shall be at least 100mW on each frequency. The output shall be measured into a 50 ohm load located at the antenna feed point and measured by a calorimetric wattmeter. For a 1000 HZ sine wave modulated at 80 percent, the RF peak power output shall be at least 100 mW on each frequency.

3.4.4.8 Beacon modulation. In the beacon mode, the carrier shall be AM with a rectangular wave and have a periodically varying audio frequency tone. The rectangular wave modulation shall provide an on/off duty cycle with an average on time duration, at the specified output power level, of from 20 to 30 percent of the cycle. The rectangular wave audio frequency tone shall periodically sweep downward over a range of at least 700 Hz, between 1600 Hz and 3000 Hz, at a rate of 2 to 3 sweeps per second. The power output level during the off time shall not exceed 15 percent of the power output level during the on time. The audio tone retrace shall not exceed five percent of the total audio tone sweep cycle time.

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3.4.5 Receiver Characteristics

3.4.5.1 Frequency Range. The receiver frequency tuning range shall be the same as that specified for the transmitter (see paragraph 3.6.3.1.1). Frequency selection shall be accomplished by the frequency select switch.

3.4.5.2 Frequency Stability and Accuracy. The receiver frequency stability and accuracy shall be the same as that specified for the transmitter see 3.4.4.2. The frequency stability of the receiver shall permit reception of a signal within +/- .005 % of the selected frequency. The input signal shall be a 30 percent modulated 1000 Hz sine wave of 10.0 hard microvolts (-93.0 dbm).

3.4.5.3 Sensitivity. The signal plus noise to noise ratio [(S+N)/N] at the output of the receiver shall be at least 6.0 dB under any combination of enviornmental conditions, see 3.5. The input signal shall be a 30 percent modulated 1000 Hz sine wave of 10.0 hard microvolts (-93.0 dbm).

3.4.5.4 Selectivity. The band pass of the receiver shall be such that the response to signals differing from the selected frequency by +/- 50 KHz shall not be more than -6.0 dB from the response to signals at the selected center frequency. At +/- 10 MHZ from the selected center frequency, the response shall be at least -55.0 dBm from the response to signals at the center frequency.

3.4.5.5 Audio output. A signal of 10.0 hard microvolts (-93.0 dBm), modulated 30 percent at 1000 Hz, applied to the receiver input terminals on either channel, shall provide sufficient audio output, to produce the required sound pressure level of at least 112 dB SPL, measured 1.0 inch from the face of the speaker with in the range of 800 to 1200 Hz.

3.4.5.6 Fidelity. Overall receiver fidelity (receiver output voltage) without speaker, in the range of 500 to 3000 Hz, shall be within (+2, -5) dB of the response to a reference frequency of 1000 Hz at an input signal level of 2000 hard microvolts (-47.0 dBm), modulated 30 percent, at any battery operating voltage.

3.4.5.7 Distortion. With the audio amplifier output required to produce the sound pressure level specified by paragraph 3.6.4.1.2, the THD of the receiver audio output voltage shall be no more than 20 percent anywhere within the range of 500 to 3000 Hz at any battery operating voltage. A test signal of 2000 hard microvolts (-47.0 dBm) modulated 30 percent, shall be applied to the receiver input terminals.

3.5 Environmental Conditions. The radio set, minus earphone, shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 1B equipment, except as modified herein. The earphone shall meet all requirements of Specification Control Drawing 12-520447.

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3.5.1 Temperature Altitude. The radio set shall provide specified performance 3.4 when subjected to the following temperature and altitude combinations:

	Operating	Non-operating
Temperature	-30°C to +55°C	-60°C to +60°C
Altitude	10,000 feet 1/	40,000 feet

1/ The speaker and microphone responses may be a maximum of 6.0 decibles (dB) less than the sea level responses.

3.5.2 Temperature Shock. The radio set nonoperating with the battery installed shall meet the performance requirements specified in 3.0 when tested in accordance with MIL-STD- 810 Method 503.3, test varation "Engineering Design", Temperature Extremes -54°C to 60°C.

3.5.3 Fungus. The radio set shall conform to Requirement 4 (fungus inert materials) of MIL-STD-454, except that the test method for uncertified equipment-shall be that required by 4.9.27.3.

3.5.4 Vibration. The radio set, nonoperating with battery installed, shall provide the specified performance after being subjected to the vibration requirements of MIL-STD-810, as specified in Figure 3.

3.5.5 Shock. The radio set with battery installed shall meet all performance requirements after being subjected to the basic design shock test of MIL-STD-810 Procedure I.

3.5.6 Salt Water Immersion. The radio set shall withstand immersion in salt water to a depth of 2.0 feet for a period of 24 hours, and to a depth of 30.0 feet for thirty minutes with no mechanical damage or degradation in performance. The radio set shall be fully functional for a minimum of 1.0 hour with the R/T immersed to a depth of 1.0 foot while the control unit and antenna remain above the surface of the water. The radio set shall not be damaged by transmitter keying while the entire assembly is underwater.

3.5.7 Salt Fog. After completion of the Salt Fog test of 4.9.27.7 the radio set and battery shall show no evidence of corrosion and meet the performance requirements as specified in 3.4.2 thru 3.4.5.7.

3.5.8 Explosion. After completion of the Explosion test of 4.9.27.8 the radio set and battery shall meet the performance requirements as specified in 3.4.2 thru 3.4.5.7.

3.5.9 Electromagnetic Interference. The radio set shall be subject the radiated susceptibility environment RS03 of MIL-STD-461 for class A1b equipment. There shall be no degradation in the performance of the R/T and control unit.

3.6 Reliability Test. A reliability test shall be included in the First Article Tests. Ten radio sets shall be tested as specified in 4.9.28.

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3.7 Talk Test. All radio sets shall be tested in accordance with the talk test specified in 4.9.29.

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, except for Requirement 5, which shall conform to MIL-STD-2000, which shall govern in the event of conflict.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items shall meet requirements of sections 3, 4 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility, of assuring that all products or supplies submitted to the Government, for acceptance, comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material. (See 6.3)

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection see 4.6

b. Initial Production Test see 4.7

c. Quality Conformance Inspection see 4.8

d. Special Tests see 4.9

4.3 Test Procedures. The procedures used for conducting Burn-in, First Article, and Quality Conformance Tests shall be prepared by the contractor for procuring activity review and acceptance (see 6.2). The right is reserved by the procuring activity, or the Government Inspector, to modify the tests, or require any additional tests deemed necessary, to determine the compliance with the requirements of this document and the contract (see 6.2). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures shall be provided and may be used when their use is acceptable to the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2).

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4.4 Rejection and Retest. When one radio set from a production run fails to meet the requirements of this document, no radio set still on hand, or produced later, shall be accepted, until the extent and cause of failure are determined and corrected. The contractor shall prepare a failure report and provide it to the procuring activity (see paragraph 4.3). The failure report shall describe the cause of failure and the action taken to preclude recurrence. After correction, the failed test(s) shall be repeated and the results reported.

4.4.1 Continuation of Tests. Individual tests may be continued at the contractor's risk, pending the investigation of a sampling test failure. Final acceptance of radio sets on hand, or produced later, shall not be made until it is determined that the equipment meets all the requirements of this document.

4.4.2 Defects in Radio Sets Already Accepted. Investigation of a test failure could indicate that defects may exist in radio sets already accepted. If so, the contractor shall provide information to the procuring activity as to all defects that are found, which units are likely to be affected (by serial number) and correction methods. Test reports shall be prepared in accordance with MIL-STD-831.

4.5 Standard Conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests: (Required field tests shall be conducted outside of buildings under ambient conditions)

a. Temperature shall be room ambient (25°C +/- 10°C).

b. Altitude shall be sea level.

c. Vibration shall be none.

d. Humidity shall be room ambient up to 90 percent relative Humidity

e. Primary power shall be a simulated battery pack set at 12.0 (+/0.1) volts DC, amperage set in accordance with paragraphs 3.4.7b and 3.4.7c.

f. Antenna impedance shall be 50 ohms Resistive (50 +/- 1 ohms)

4.6 First Article Inspection. First Article Inspection shall consist of all inspections, tests, and examinations listed in Table I and shall be made on equipment representative of the production equipments to be supplied under the contract. First Article Inspection shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so

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desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. No First Article Inspections shall be conducted prior to approval of First Article Test procedures by the procuring activity. (See 6.3)

4.6.1 First Article Test Data. The contractor shall make available all data collected in conducting these tests for procuring activity review and approval (see 4.3). This data, including a list of all electrical and electronic parts, giving their specified voltages and ambient surface temperatures, shall be made available on first production contracts. The ambient and surface temperatures shall be obtained under the extreme high temperature operating condition.

4.7 Initial Production Tests. When required by contract, one of the first ten production radio sets shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This radio set shall be selected by the procuring activity after the equipment has successfully completed burn-in and all individual tests. No other tests shall be conducted on the radio set prior to starting the initial production tests. The First Article samples shall not be selected for this test.

4.7.1 Scope of Tests. The radio set shall be subjected to all tests the procuring activity deems necessary to assure that production equipment is equivalent to the previously approved First Article samples in design, construction, workmanship, performance and quality, and that it meets all applicable requirements.

4.7.2 Initial Production Sample Approval. Approval of the initial production sample shall be made by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be implemented on equipment previously accepted, at no cost to the Government, when requested by the procuring activity.

4.7.3 Reconditioning of Initial Production Test Sample. The initial production test sample shall be reworked by the contractor, by replacing all worn or damaged items, after completion of the initial production tests. After rework, the contractor shall resubmit the equipment for acceptance.

4.8 Quality Conformance Inspection. The contractor shall furnish all samples, and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare test reports showing quantitative results for all quality conformance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Quality conformance inspection shall consist of the tests listed in Table I.

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TABLE I. REQUIREMENTS AND INSPECTIONS

	REQUIREMENT/ PARAGRAPH	INSPECTION PARAGRAPH
Dimensions	3.3.5	4.9.1
Weight	3.3.6	4.9.2
Power Source	3.3.7	4.9.3
Battery Reversal	3.3.7.1	4.9.4
Battery Operating Voltage and Current	3.3.7.2	4.9.5
Manual Operating Controls	3.3.8	4.9.6
Umbilical Cable	3.3.10	4.9.7
Earphone Charactistics	3.3.11	4.9.8
Stress Screening (Burn in)	3.4.1	4.9.9
Sidetone	3.4.2	4.9.10
Warm-up Time	3.4.3	4.9.11
Frequency Range	3.4.4.1	4.9.12
Frequency Stability and Accuracy	3.4.4.2	4.9.13
Microphone Sensitivity	3.4.4.3	4.9.14
Modulation	3.4.4.4	4.9.15
Distortion	3.4.4.5	4.9.16
Fidelity	3.4.4.6	4.9.17
RF Power Output	3.4.4.7	4.9.18
Beacon Modulation	3.4.4.8	4.9.19

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TABLE I. REQUIREMENTS AND INSPECTIONS Cont'd

	REQUIREMENT/ PARAGRAPH	INSPECTION PARAGRAPH
Frequency Range	3.4.5.1	4.9.20
Frequency Stability and Accuracy	3.4.5.2	4.9.21
Sensitivity	3.4.5.3	4.9.22
Selectivity	3.4.5.4	4.9.23
Audio output	3.4.5.5	4.9.24
Fidelity	3.4.5.6	4.9.25
Distortion	3.4.5.7	4.9.26
Temperature Altitude	3.5.1	4.9.27.1
Temperature Shock	3.5.2	4.9.27.2
Fungus	3.5.3	4.9.27.3
Vibration	3.5.4	4.9.27.4
Shock	3.5.5	4.9.27.5
Salt Water Immersion	3.5.6	4.9.27.6
Salt Fog	3.5.7	4.9.27.7
Explosion	3.5.8	4.9.27.8
Electromagnetic Interference	3.5.9	4.9.27.9
Reliability Test	3.6	4.9.28
Talk test	3.7	4.9.29
Workmanship	3.8	4.9.30

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4.9 Methods of inspection

4.9.1 Dimensions. The radio set shall be inspected for conformance to the dimensions specified in 3.3.5.

4.9.2 Weight. The radio set shall be inspected for conformance to the weight specified in 3.3.6.

4.9.3 Power Source. The radio set shall be inspected for conformance to the power source requirement specified in 3.3.7.

4.9.4 Battery Reversal. The radio set shall be inspected for conformance to the battery reversal requirement specified in 3.3.7.1.

4.9.5 Battery Operating Voltage and Current. The radio set shall be inspected to insure that the battery operating voltage and current are in conformance with 3.3.7.2.

4.9.6 Manual Operating Controls. The radio set shall be inspected to insure that all controls operate in conformance with 3.3.8.

4.9.7 Umbilical cable. The umbilical cable shall be inspected to insure conformance with 3.3.10.

4.9.8 Earphone Characteristics. Earphones shall be inspected in accordance with Specification Control Drawing 12-520447.

4.9.9 Stress Screening (Burn-in). Each radio set shall be subjected to a stress screening (burn-in) test consisting of a temperature and a vibration test. The temperature test shall be in accordance with Figure 4. During the hot portion of each temperature cycle the radio set shall be subject to two ten minute periods of random vibration in accordance with Figure 3. It should be noted that random vibration and temperature cycling shall occur simultaneously. The duty cycle shall be a 30 minute cycle with 27 minutes in receive mode and three minutes in transmit mode. One-half of the radio sets in the burn-in lot will have 243.0 MHZ selected and the other one-half will have 282.8 MHZ selected. The temperature cycle shall be based on a thermal survey conducted in accordance with MIL-STD-781. A minimum of two thermal cycles shall be required. Temperature limits are specified as follows:

a. Lower Limit -60°C

b. Upper Limit +60°C

Proper operation of each radio set shall be verified at -30°C and at +55°C. Any failure occuring during the burn-in period shall be considered non-relevant but shall be reported in appropriate reliability test reports. After each repair, the unit shall be operated for at least four additional failure-free hours.

4.9.10 Sidetone. The radio set shall be inspected to determine that the sidetone is in conformance with 3.4.2.

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4.9.11 Warm-up Time. The radio set shall be inspected to determine the capability of the equipment to conform to the warm-up time requirement of paragraph 3.4.3.

4.9.12 Frequency Range. The radio set shall be inspected to determine conformance with the frequency range requirements of 3.4.4.1.

4.9.13 Frequency Stability and Accuracy. The radio set shall be inspected to determine conformance with the frequency stability and accuracy requirements of 3.4.4.2.

4.9.14 Microphone Sensitivity. The sensitivity of the microphone shall be inspected to determine conformance with 3.4.4.3.

4.9.15 Modulation. The radio set shall be inspected to determine conformance with the modulation requirements of 3.4.4.4.

4.9.16 Distortion. The radio set shall be inspected to determine conformance with the distortion requirements of 3.4.4.5.

4.9.17 Fidelity. The radio set shall be inspected to determine conformance with the fidelity requirements of 3.4.4.6.

4.9.18 RF Power Output. The radio set shall be inspected to determine conformance with the RF Power output requirements of 3.4.4.7.

4.9.19 Beacon Modulation. The radio set shall be inspected to determine conformance with the beacon modulation requirements of 3.4.4.8.

4.9.20 Frequency Range. The radio set shall be inspected to determine conformance with the frequency range requirements of 3.4.5.1.

4.9.21 Frequency Stability and Accuracy. The radio set shall be inspected to determine conformance with the frequency stability and accuracy requirements of 3.4.5.2.

4.9.22 Sensitivity. The radio set shall be inspected to determine conformance with the sensitivity requirements of 3.4.5.3.

4.9.23 Selectivity. The radio set shall be inspected to determine conformance with the selectivity requirements of 3.4.5.4.

4.9.24 Audio Output. The radio set shall be inspected to determine conformance with the audio output requirements of 3.4.5.5.

4.9.25 Fidelity. The radio set shall be inspected to determine conformance with the fidelity requirements of 3.4.5.6.

4.9.26 Distortion. The radio set shall be inspected to determine conformance with the distortion requirements of 3.4.5.7.

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4.9.27 Environmental Tests. A sample of radio sets shall be subjected to environmental tests, in accordance with MIL-STD-810 for Class I equipment, with the additions and exceptions specified herein. When the two documents conflict, the requirements stated herein shall govern. Each environmental test shall be followed by the electrical performance tests specified in 3.4.2 thru 3.4.5.7.

4.9.27.1 Temperature-Altitude. The radio set shall be subjected to temperature-altitude tests in accordance with MIL-STD-810 procedure III

4.9.27.2 Temperature Shock. The radio set shall be subjected to temperature shock test in accordance with MIL-STD-810.

4.9.27.3 Fungus. Equipment certified as having fungus inert material in all components need not be subjected to the fungus resistance test. Uncertified equipment shall be subjected to fungus test in accordance with MIL-STD-810.

4.9.27.4 Vibration. The radio set shall be subjected to a vibration test in accordance with MIL-STD-810, Curve IVa, non-operating, with battery installed.

4.9.27.5 Shock. The radio set shall be subjected to the basic design shock test in accordance with MIL-STD-810 and 3.5.5.

4.9.27.6 Salt Water Immersion. The radio set shall be immersed in salt water to ensure conformance with requirements paragraph 3.5.6. The salt water solution shall be in accordance with ASTM D1141-52.

4.9.27.7 Salt Fog. The radio set shall be subjected to a salt fog test in accordance with MIL-STD-810.

4.9.27.8 Explosion. The radio set, including battery, shall be subjected to Procedure I, Explosion Test, of MIL-STD-810.

4.9.27.9 Electromagnetic Interference. The radio set shall be subjected to the Radiated Susceptibility Test, Procedure RS03 of MIL-STD-462, to determine the capability of the equipment to conform to the requirements of paragraph 3.5.9.

4.9.28 Reliability Test. A reliability test shall be included in the First Article Tests. Ten radio sets shall be tested and shall, at a minimum, be subjected to a 10 minute random vibration period in accordance with Figure 5 during each thermal cycle of Figure 6. It should be noted that vibration and temperature cycling shall occur simultaneously. The temperature cycle shall be based on a thermal survey conducted in accordance with MIL-STD-781. Temperature limits are specified as follows:

a. Lower Limit -60°C.

b. Upper Limit +60°C.

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Proper operation of each radio set is to be verified at -30°C and +55°C. Each radio set shall have successfully completed the burn-in requirement of 4.9.9 prior to beginning the reliability qualification test. The accept-reject criteria imposed shall be that of MIL-STD-781, Test Plan IIIC, with the lower test MTBF specified in paragraph 3.4.1.1.

4.9.28.1 Reliability Assurance Tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Every production lot shall be subjected to reliability sampling tests. (A production lot is defined as the number of radios produced in a calendar month.)

4.9.28.2 Reliability Sampling Tests. Reliability sampling testing shall be conducted on a sample of each production lot (10 percent of the lot, up to 20 units). Vibration, temperature limits, proper operation, and the burn-in requirements shall be in accordance with 4.10.8 herein. The accept-reject criteria imposed shall be that of Test Plan IVC, MIL-STD-781, with the lower test MTBF specified in paragraph 3.3.3.1 herein.

4.9.28.3 Test Details. The reliability assurance test procedures prepared for this equipment shall include the following:

a. Length of the Test Cycle

b. Heat Portion of the Test cycle

c. Performance Characteristics to be Measured

d. Special Failure Criteria and Preventive Maintenance to be allowed during the test

Test procedures shall be approved by the procuring activity prior to the beginning of the First Article test phase of the reliability assurance tests (see paragraph 4.3). The procuring activity may stop the acceptance of equipment when a reject situation exists. Testing shall not resume until the cause of the reject is corrected by the contractor. Failure classification shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.9.29 Talk Test. All radio sets shall be used to "talk" to UHF radios capable of transmitting and receiving AM voice transmission on 243.0 MHz and 282.8 MHz. Proper operation of transmitter sidetone shall be verified on both channels specified in 3.4.2.

4.9.30 Workmanship. The radio set shall be inspected for conformance to the workmanship requirement specified in 3.8.6.

4.9.31 Special Tests. Special tests shall be conducted, when specified in the contract or purchase order, on a quantity of equipment for the purpose of checking the effect of any design, or material change, on the performance or supportability of the radio set, (see 3.3.4) and to ensure adequate quality control. The equipment selected for special tests, may be selected from equipment previously subjected to the sampling or reliability assurance tests.

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4.9.31.1 Special Test Schedule. Selection of radios for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required, as determined by the procuring activity.

4.9.31.2 Scope of Tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval, prior to conducting the tests.

4.10 Equipment Failure. Should a failure occur during either First Article, acceptance, reliability assurance, or special tests, the following actions shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare and make available proposed corrective action, acceptable to the procuring activity, intended to reduce the possibility of the same failure(s) occurring in future tests (see 4.3).

d. Where practical, include a test in the individual tests to check all equipment for this requirement until assurance has been obtained that the defect has been corrected.

4.10.1 Failure Classification. All incidents occurring during reliability assurance test shall be classified and reported, as either relevant or non relevant, in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept/reject decision.

4.10.2 Relevant Failures. All test incidents shall be considered as relevant failures, unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

4.10.3 Non-relevant Failures. Although non-relevant failures are not used for reliability assurance test accept/reject decisions, all test article incidents and corrected equipment failures shall be recorded and reported (see 4.10).

5.0 PACKAGING

5.2 Preservation. Unless otherwise specified in the contract or purchase order, preservation shall be Level A in accordance with MIL-STD-794.

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5.3 Packaging. Unless otherwise specified in the contract or purchase order, packaging shall be Level A in accordance with MIL-STD-794.

5.4 Packing. Unless otherwise specified in the contract or purchase order, packing shall be Level C, in accordance with MIL-STD-794, for CONUS shipments; and Level A for X-CONUS shipments.

5.5 Marking. In addition to any special marking required by the contract, each unit package, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

6.0 NOTES

6.1 Item Description. The AN/PRC-125 Radio Set (Figure 1) is a dual channel, battery powered radio set providing communications between a Navy rescue swimmer and the rescue aircraft. The radio set provides two-way communications on 243.0 MHz and 282.8 MHz and transmits a swept tone at 243.0 MHz in the beacon mode. The rescue swimmer will carry the radio set in the LPU-28/P (Figure 2) life vest, with the Receiver/Transmitter (R/T) unit attached to the lower part of the life vest in the pouch. The AN/PRC-125 shall allow hands off operation when the control unit is attached to the upper portion of his LPU-28/P life vest with hook and pile material. The control unit/antenna assembly is connected to the R/T unit by a three foot watertight cable. The AN/PRC-125 is designed to operate normally while submerged in salt water, and suffer no damage, after immersion in salt water, to a depth of 30 feet for a period of thirty minutes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. The latest revision of this document.

b. When a first article is required for inspection and approval see 4.6.

c. Initial production tests.

d. Additional special tests, if required see 4.9.31.

e. Additional requirements for test procedures (paragraph 4.3).

f. Selection of applicable levels of packaging and packing see 5.1.

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423 must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.1.1	DI-NDTI-80809A	Inspection/Test Reports	10.27 only
4.6	" "	" " "	Use contractor format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must6 be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First Article. When a First Article inspection is required, the items tested shall be First Article samples. The procuring activity should include specific instructions in acquisition documents regarding arrangements for examinations, approval of First Article test results, and disposition of First Article samples. Note: Data item descriptions related to this document and identified in the CDRL, should be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractor, in connection with specific acquisition functions, should be obtained from the Naval Publications and Forms Center, or as directed by the contracting officer.

6.5 Government Furnished Equipment (GFE). If required by contract, the following equipment will be furnished to the contractor, by the Government, for use in connection with the AN/PRC-125 procurement contract:

a. Quantity - 1 each, SDU5/E Strobe Light

b. Quantity - 1 each, Model LPU-28/P Life Vest

6.6 Subject term (key word) listing.

Battery powered radio
Rescue aircraft
Rescue radio

Preparing activity:
Navy-AS
(Project 5821-N246)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-85698(AS)	19 March 1993

3. DOCUMENT TITLE

RADIO SET, AN/PRC-123 (NAVY SWIMMER RADIO)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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